

CS & IT ENGINEERING

Algorithms

Algorithms



DPP 03 Discussion Notes

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#Q. Consider an array of n elements. If we apply the linear search algorithm to perform a searching operation on the array, in a worst-case and average case scenario, it takes p and q comparisons and the value of $p + q$ is 41. Then the number of elements present in the array is 27?

Linear search.



no of comparison in worst case = $n = p$

min 1 4

in Average case =

$$\frac{1+2+3+\dots+n}{n} = \frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2} = 4$$

$$p+q = 41$$

$$n + \frac{n+1}{2} = 41$$

$$3n+1 = 82$$

$$3n = 81 \quad n = 27$$

[MCQ]



#Q. What is the time and space complexity of the binary search algorithm?

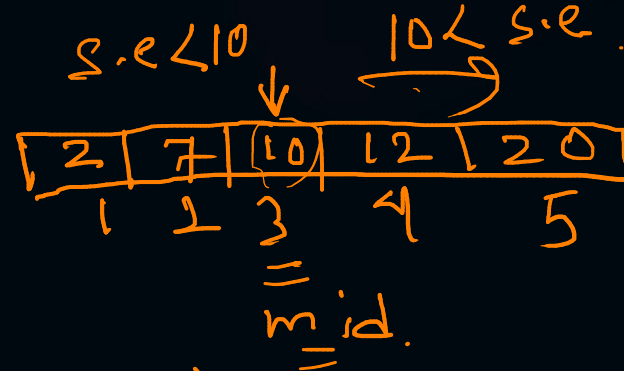
A $O(n), O(\log n)$

B $O(\log n), O(n)$

C $O(n), O(n)$

D $O(\log n), O(\log n)$

B.S.



$$T(n) = T(n/2) + C$$

$$= T(n/2^2) + C + C$$

$$= T(n/2^2) + 2C$$

$$= T(n/2^3) + 3C$$

! after k terms

$$= T(n/2^k) + kC$$

assume $\frac{n}{2^k} = 1$

$$k = \log_2 n$$

↓

$$\therefore \Rightarrow O(\log n)$$

$$\begin{aligned} T(n) &= T(1) + C \cdot \log n \\ &= O(\log n) \end{aligned}$$



#Q. Consider the following array

Array	5	3	55	82	40	25	15	30
-------	---	---	----	----	----	----	----	----

How many swaps required to sort the above array using bubble sort 13?

pass1:

5	8	55	82	90	25	15	30
3	5		40	82	82	82	82
				25	15	30	

Comp
7

Swap
5

pass 2: (3, 5, ~~55~~, ~~40~~, ~~35~~, ~~15~~, ~~30~~) 82

40	55	55	55	55
	25	15	30	

6 4.

pass 3: (3, 5, ~~40~~, 25, ~~15~~, ~~30~~) 55, 82

5 3

pass 4: $(3, 5, \overset{15}{25}, 15, 30)$ 40 55 82

25

4



pass 5: 3, 5, 15, 25, 30, 40, 55, 82

7

3	0
2	0
1	0
	13

[MCQ]



#Q. Consider the following array

Array	3	24	15	232	130	780	1024	512
-------	---	----	----	-----	-----	-----	------	-----

what will be 7th element after 3rd pass of Radix sort ?

4 digit.
↓
4 pass.

A

780

B

232

C

512

D

130

0003, 0024, 0015, 0232, 0130, 0780, 1024, 0512
↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑
0780 0512 1024 0015
1st pass 0/30 2/32 3/03 4/24 5/15 6 7 8 9
0/p → 0130, 0780, 0232, 0512, 0003, 0024, 1024, 0015
 ↑ ↑ ↑ ↑ ↑ ↑ ↑

2nd pass

$\frac{0003}{0}$	$\frac{0015}{0512}$	$\frac{1024}{0024}$	$\frac{0130}{0232}$	4	5	6	7	$\frac{0780}{8}$	9
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O/p $\rightarrow$ 0003, 0512, 0015, 0024, 1024, 0130, 0232, 0780

3rd pass

$\frac{1024}{0024}$	$\frac{0015}{0003}$	$\frac{0130}{1}$	$\frac{0232}{2}$	3	4	$\frac{0512}{5}$	6	$\frac{0780}{7}$	8	9
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O/p $\rightarrow$ 0003, 0015, 0024, 1024, 0130, 0232, 0512, 0780

4th pass

$\frac{0780}{0512}$	$\frac{1024}{0003}$
0232	
0130	
0024	
0015	
0003	
0	

1 2 3 4 5 6 7 8 9

Sorted.

4th pass O/p $\rightarrow$ 3, 15, 24, 130, 232, 512, 780, 1024

[MCQ]

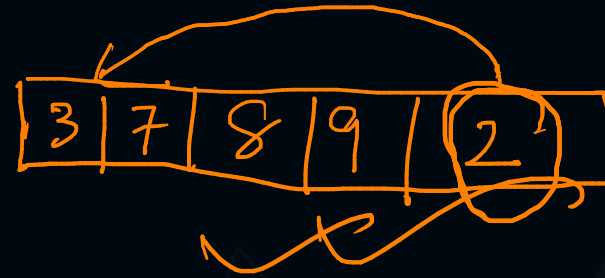


#Q. Consider an array of n elements where almost all the elements are sorted, rather than one or two elements. Now, if we want to sort the given array, then which of the following sorting algorithms will be efficient?

A

Merge sort

$n \log n$ (BC, AC)
WC



B

Quick sort

$n \log n$ (BC, AC)
 n^2 (WC) $\Rightarrow$ array element sorted or almost sorted.

C

Selection sort

$\Rightarrow O(n^2)$

D

Insertion sort

$O(n)$ (BC) $\Rightarrow$ array element sorted or almost sorted.
 $O(n^2)$ (WC, AC)

#Q. Which of the following options is/are correct ?

A

Space complexity of Counting sort is $O(k)$, where k is the maximum element size in the given input array.

B

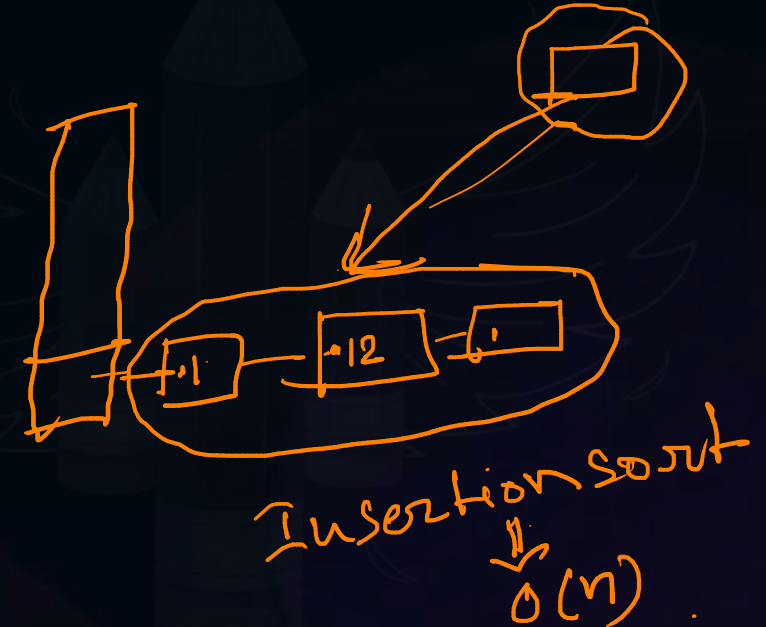
In Bucket sort more than one element belong to same bucket then we use Selection sort to sort them efficiently.

C

Bucket sort worst case time complexity is $O(n^2)$

D

Bubble sort is a in place sorting technique.



Ans A, C, D
 $O(1)$
inplace

$1 \leq S \leq \log n$
inplace

Counting Sort

out
Sorting.

100 size
array

↓
out

given array $\rightarrow$ 1, 2, 2, 100, 3

5 ele.

$\Rightarrow$ 1, 2, 2, 3, 100 $\rightarrow$ sort.

1	1
2	2
3	1
...	
100	1

[MCQ]



#Q. Consider an array A

A

3	5	80	90	2	65	1	10	12
---	---	----	----	---	----	---	----	----

If we apply selection sort on the above array then what is the output after ~~3rd~~ ^{4th} pass?

A 1, 2, 3, 5, 65, 90, 80, 10, 12

B 1, 2, 3, 5, 90, 65, 80, 10, 12

C 1, 2, 3, 5, 90, 65, 10, 80, 12

D 1, 2, 3, 5, 12, 65, 80, 90, 10

Ans B

pass1: 1 ~~3~~, (5, 80, 90, 2, 65, ³~~1~~, 10, 12) $\Rightarrow$ 1st min

pass2: 1, 2 ~~3~~, (80, 90, ⁵~~2~~, 65, 3, 10, 12) $\Rightarrow$ 2nd min.

pass3: 1, 2, 80, (90, 5, 65, ³~~3~~, 10, 12)

pass4: 1, 2, 3, 90, (5, ⁹⁰~~5~~, 65, 80, 10, 12)

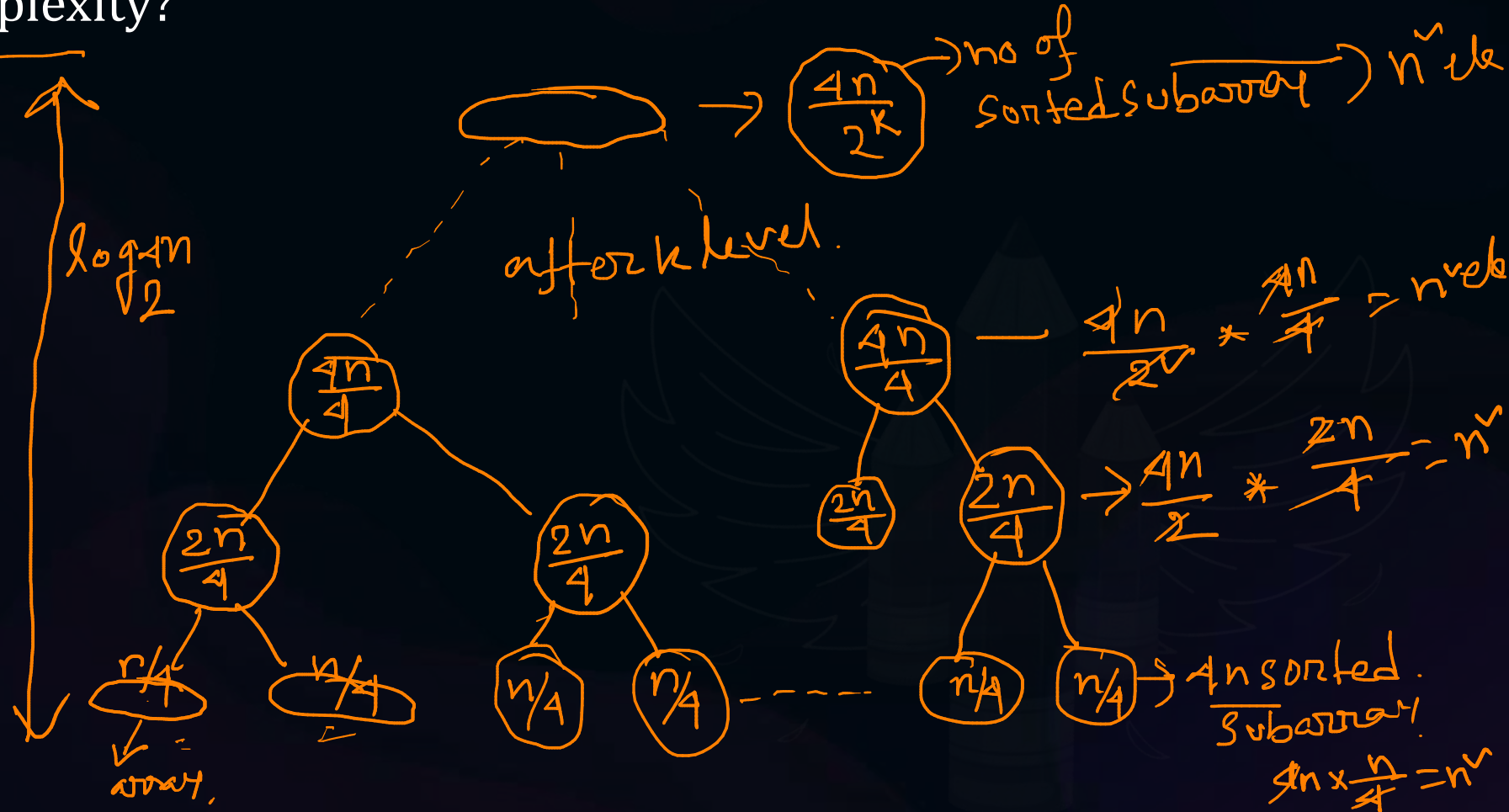
1, 2, 3, 5, 90, 65, 80, 10, 12 $\Rightarrow$ pass4 o/p.

[MCQ]



#Q. Consider there are $4n$ sorted subarrays where each subarray size is $\frac{n}{4}$. Now if we merge them into a single sorted array then in the worst case what will be the time complexity?

- A** $\theta(n \log n)$
- B** $\theta(n^3 \log n)$
- C** $\theta(n^2 \log n)$
- D** $\theta(\sqrt{n} \log n)$



$$\left. \begin{aligned} \frac{4n}{2^k} &= 1 \\ 2^k &= 4n \\ k &= \log_2 4n \end{aligned} \right\}$$

Each level $n^{\vee}$ ele is moving.

like that happened for $\log_2 4n$ level.

$$TC = n^{\vee} \times \log_2 4n = \Theta(n^{\vee} \log n)$$

[MCQ]



#Q. Which of the following recurrence relations represents the Quick Sort worst-case time complexity scenario?

A

$$T(n) = 2T(n/2) + O(\log n)$$

B

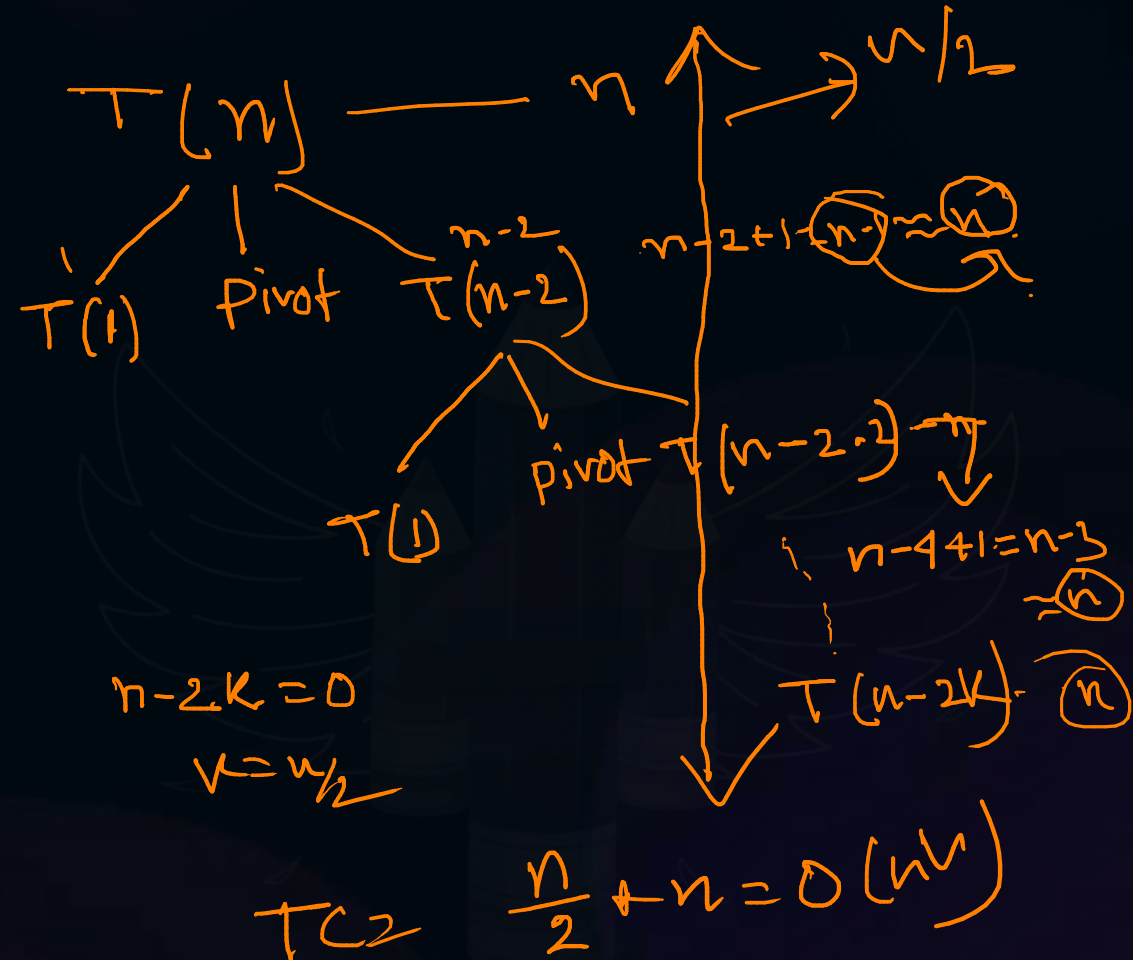
$$T(n) = 2T(n/2) + O(n)$$

C

$$T(n) = T(n-2) + T(1) + O(n)$$

D

$$T(n) = T(n-2) + T(1) + O(\log n)$$



Quick sort worst case general recurrence relation.



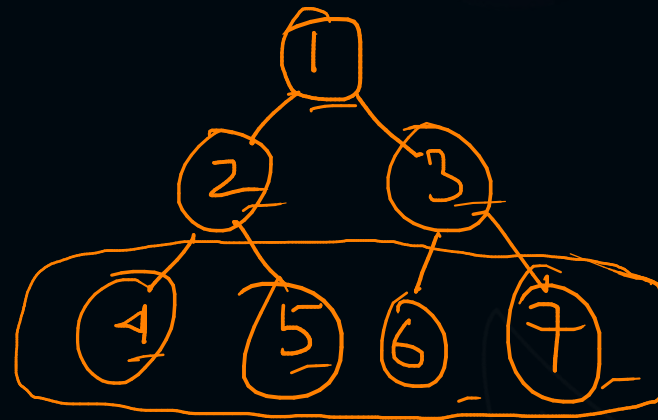
$$\Rightarrow T(n) = T(\underline{c-1}) + T(\underline{n-c}) + \underline{O(n)} \quad , \quad c \text{ integer.}$$

[MCQ]



#Q. Assume a Min-heap tree containing n^2 elements then time required to find out the maximum element in this tree?

- A** $\theta(1)$
- B** $\theta(\log n)$
- ~~**C** $\theta(n^2)$~~
- D** $\theta(n^2 \log n)$



Total nodes = 7.
leaf nodes = $\lceil \frac{7}{2} \rceil$

$$\text{no of leaf nodes} = \lceil \frac{n^2}{2} \rceil = \frac{n^2}{2}$$

$\frac{n^2}{2}$ ele to get maximum $\Rightarrow$ bubble sort first pass
 $\Downarrow$
Maximum ele.

$$\theta(n^2) = \left(\frac{n^2}{2} - 1 \right) + \left(\frac{n^2}{2} - 1 \right)$$

[MCQ]



#Q. Consider the array elements:

10, 2, 7, 15, 9, 20, 23

Develop a max heap tree where we insert all the elements one at a time. Then, which one of the following options represent the correct sequence of max heap tree?

A

23, 10, 20, 2, 9, 7, 15

B

23, 15, 20, 2, 9, 10, 7

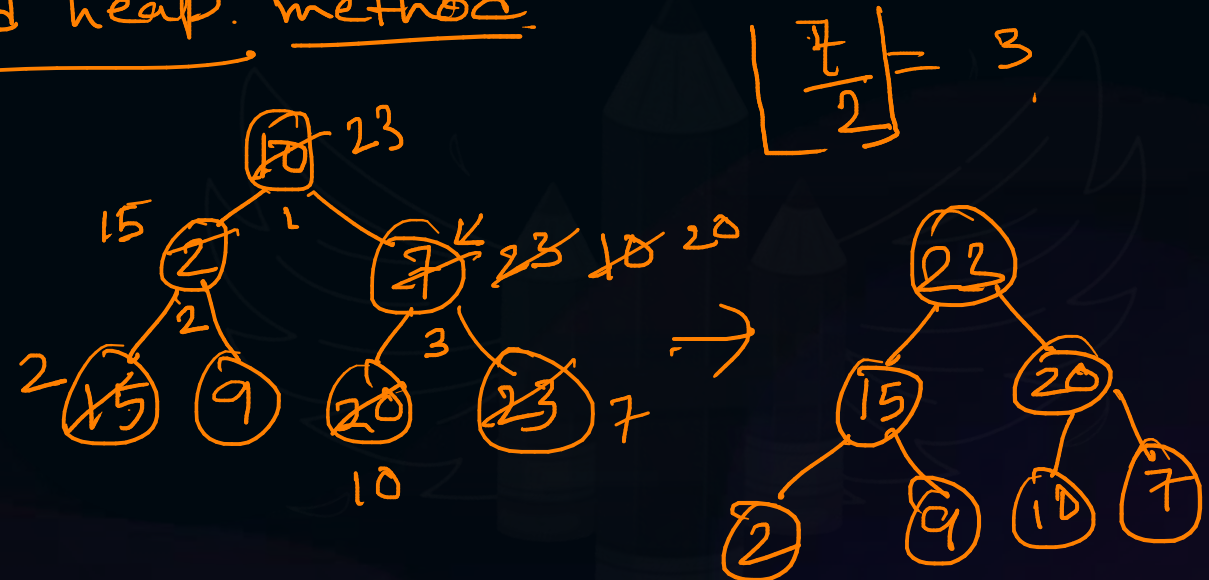
C

23, 15, 20, 9, 2, 10, 7

D

23, 15, 20, 2, 9, 7, 10

Build heap. method.



23, 15, 20, 2, 9, 10, 7.

[MSQ]



#Q. Which of the following statements is/are not false? true

- ☒ **A** If we decrease an element of a min-heap tree of n elements, then the heapify purpose it may take $O(\log n)$ time.
- ☒ **B** Quick sort and Merge sort are both stable algorithms.
Not stable.
- ☒ **C** Quick sort, Merge sort, and Heap sort all have the same worst and average case time complexity.
- ☒ **D** If we apply Quick sort on a sorted array of n elements, then it takes $O(n^2)$ time.

Ans D, D

Sorted
↓
unbalanced tree.
↓
 $O(n^2)$

Quick sort

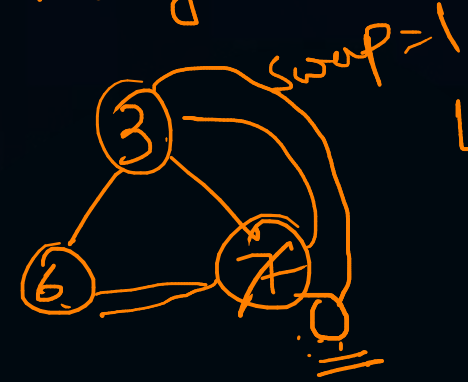


→ 10a, 10b, 20
 → 10b, 10a, 20
 ↓
not stable.

Quick sort
 Merge sort
 heap sort.



⇒ $\log n$ $\leftrightarrow$ comp + swaps.
 $\rightarrow 2 \log n + \log n = \Theta(\log n)$



$1 + 1 = 2 \text{ comp.}$

Average case

$n \log n$.
 $n \log n$
 $n \log n$

worst case

n^2
 $n \log n$.
 $n \log n$





THANK - YOU